SpaceTformer: a Spacetime Transformer for Photon Energy Reconstruction in the LHCb PicoCal under Pileup

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Abstract

We study photon energy regression in the proposed LHCb Upgrade II PicoCal electromagnetic calorimeter on a simulated minimum-bias sample, where the photon's median share of the energy in its own cluster is 19% in the innermost region. SpaceTformer, a token transformer over calorimeter cells — each token a point in detector space and time — with a physics-constrained readout, reaches an aggregate relative resolution $\sigma_{\text{eff}} = 0.0388 \pm 0.0002$, measured fully out-of-sample under a complete ten-fold cross-validation protocol (every event predicted by models that never saw it). Scored against a six-feature gradient-boosted reference on the events the two share, the improvement is a factor 3.15 ± 0.05, with the tree behind in every one of 400 paired resamples and in every region separately. In a paired comparison with the encoder as the only varied component, it outperforms re-implementations of ParticleNet and GravNet in every measured cell at every seed, at 1.8–3.6× their training speed — a margin that survives a per-baseline hyperparameter sweep. The main contribution is a diagnosis: the resolution ceiling was set by the input window, not the model. A seed-centred 9×9 window already holds 99.8% of the photon, so the width was not buying shower coverage: it was buying robustness to a window centre that pileup displaces by three cells, and above that, cells that sample the pileup field rather than the photon. The starting window held under half of the innermost region's cluster energy, and fixing the window's size and centring cut the worst region–energy bin by 53% (0.165 → 0.078), while ten encoder families, four gate-supervision protocols, five timing constructions, a capacity sweep and a recipe sweep all measured null against the same bins. Two mechanisms move the number at the per-cent level: an auxiliary head that regresses the photon's position alongside the energy, and a learned two-stage window built on it — a first pass points at the photon (median error 0.16 cells), the window is re-cut around it, a second pass measures. On the development protocol the two-stage ensemble reads 0.0378 against 0.0388, better in every one of 400 paired resamples; under cross-validation the same construction does not reproduce that gain, while the auxiliary head is worth about 1% under both protocols. We report both measurements. A directly measured data-scaling curve, $\sigma_{\text{eff}} \propto N^{-0.18}$ over four within-protocol subsample points, predicts that tripling the simulated sample would bring the aggregate to ≈ 0.033, leaving training data as the only open lever. The tripled-sample value is an extrapolation beyond the measured range and is labelled as such.

1 Introduction

The PicoCal is the proposed electromagnetic calorimeter for LHCb Upgrade II [20]. It is a sampling calorimeter built from 12×12 cm modules of two technologies — a spaghetti calorimeter close to the beam and shashlik further out — segmented so that the cell pitch grows with distance from the beam pipe: five regions at 15, 30, 40, 60 and 120 mm, from one to 64 cells per module. Every cell is read out at both depths and carries a timestamp. The design intent of that segmentation is to keep the occupancy per cell manageable where the particle flux is highest, which is where the beam pipe is.

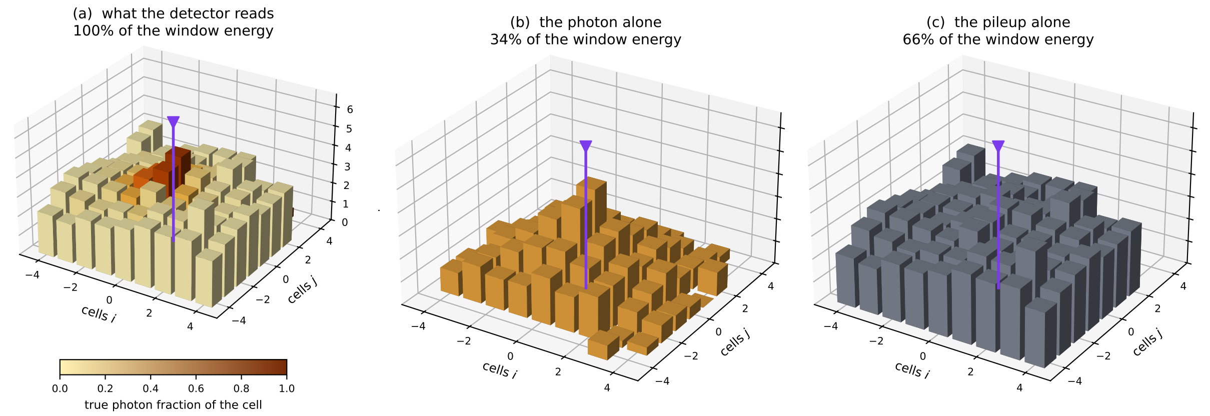

Figure 1: The separation problem, on one cluster of a paired sample where the photon's contribution to every cell is known (15 mm region, $E_\gamma = 48\,\mathrm{GeV}$). (a) What the detector reads: bar height is the cell energy on a logarithmic scale, colour is the fraction of that cell which belongs to the photon. (b) The photon's contribution alone and (c) the pileup contribution alone, on the same scale. The photon supplies a third of the window energy and shares its cells with the pileup; no cut on position or on energy separates them, which is why the readout of Section 4 weights cells rather than selecting them. The arrow marks the true photon entry point.

It also sets the shape of the problem this paper is about. At the luminosities Upgrade II targets [state the simulated ν / luminosity — ask the collaboration], every photon cluster carries energy from tens of simultaneous proton–proton interactions. In the sample used here the photon's median share of the energy inside its own cluster is 58% over all regions but only 19% at 15 mm and 32% at 30 mm. The gradient follows the beam pipe, not the cell size: the inner regions are contaminated because the particle flux there is highest, and the fine segmentation is the response to that flux rather than its cause — a coarser inner calorimeter would be contaminated more, not less. Reconstructing the photon energy in those regions then stops being a calibration problem and becomes a separation problem. Figure 1 makes the statement literal: on a paired sample in which the photon's own deposit is known cell by cell, two thirds of the energy inside the window belongs to other interactions, and it is spread over the same cells as the photon rather than sitting beside them.

Machine-learned regression is the natural tool, and the field has produced a family of increasingly sophisticated architectures for calorimeter and jet tasks — graph networks such as ParticleNet [1] and GravNet [2], attention models [4], and hybrids. One of them addresses this detector and this task directly: a graph network for energy reconstruction in the Upgrade II electromagnetic calorimeter [18], built on an attention-enhanced GarNet and compressed for real-time use. We could not compare against its numbers, which are not published, so what we can say is narrower and we say only that: the GarNet family's closest published relative, GravNet, was re-implemented here behind an identical readout, loss, splits and window, and reads 0.0437 against the transformer's 0.0399 as single models on the same events. That is a statement about the architecture in our pipeline, not about their result. Papers in this family usually compare against numbers published on other datasets, which makes it hard to know how much of a reported gain comes from the architecture and how much from everything around it. We instead fix one training protocol — loss, splits, seeds, readout, calibration — and vary exactly one component at a time, treating every proposed improvement as a hypothesis to be measured. One hundred and thirty-four distinct configurations were measured this way, each with a saved per-event prediction set and cross-validated arms counted once (the complete scored ledger is versioned with the code).

The thesis this discipline produced can be stated in one sentence: *on pileup-dominated*

PicoCal data, the resolution ceiling was set by the input window, not the model — widening it until it tolerated the centring error, and then improving the centre, cut the worst bin by 53%, while every architectural alternative measured null or, in the one case that attacked the window itself, did not reproduce across protocols (Section 5.3), leaving training-set size as the dominant open lever. The paper presents the primary result and its uncertainty protocol (Section 5.1), the diagnosis with its mechanism measurements (Sections 5.2–5.4), a quantitative account of what now limits the measurement (Section 5.5), the controlled architecture campaign (Section 5.6), the record of measured negative results (Sections 5.7–5.9), deployment cost (Section 5.10), and the scaling case (Section 5.11).

2 Related work

Graph networks for irregular detectors. Point-cloud learning entered vision through PointNet [21] and EdgeConv [22], and particle physics through ParticleNet [1] (EdgeConv over dynamically recomputed k-nearest-neighbour graphs) and GravNet [2] (aggregation weighted by distance in a learned latent space, designed for calorimeters); the same lineage now powers full particle-flow reconstruction (MLPF [23]) and symmetry-equivariant taggers (LorentzNet [24]). ParticleNet and GravNet, the two most often recommended for our task, are re-implemented and measured here under a paired protocol (Section 5.6). We read the outcome not as a criticism of the architectures but as evidence that at $O(10^2)$ tokens with severe per-cell noise, local graph connectivity provides no advantage over full attention.

Attention in particle physics. The transformer [3] entered jet physics through the Particle Transformer [4], whose pairwise physics-motivated attention bias is the direct ancestor of one of our losing arms. TimeSformer [5] showed that factorising attention along the time axis is effective for video; our time-sliced attention arm transplants that idea using per-cell time as the axis, and it loses — video frames are dense and redundant along time, calorimeter timestamps are one noisy scalar per token. Patch-hierarchical attention for jets [6] cuts attention cost by grouping tokens into patches — there consecutive chunks of a particle list under a fixed sequence ordering (k_{T} in their experiments), explicitly not spatial regions. We transplanted the cost argument to a geometry where the hierarchy can be spatial: individual cells in the core, coarse ring summaries outside. It is the one transplanted idea that survives measurement. ClusTEX [7] separates global from local detector coordinates; the same separation measured here is neutral.

Set, flow, and mixing methods. Deep Sets [8] pool per-particle embeddings additively; Energy Flow Networks [9] restrict that pooling to an energy-weighted sum over angular directions, which is what makes them infrared- and collinear-safe — the unrestricted Deep Sets form is not. Our EFN arm confirms that a linear pooling of this kind is too weak when the task is subtractive — separating overlapping energy fields rather than summarising one. MLP-Mixer-style token mixing [10] replaces attention with a learned but input-independent MLP across tokens; a masked Mixer arm measured here loses heavily (Section 5.6) — under pileup the informative cells move event by event, which content-based attention tracks and a fixed mixing pattern weights cannot.

Linear attention and state-space models. The Mamba lineage [11] offers linear-complexity sequence mixing whose advantage is most pronounced at long sequence length (its published comparisons run to thousands of tokens and beyond). Our events carry 81–289 tokens, where full attention is exact and inexpensive, so we deprioritised the family without measurement and say so explicitly.

Self-supervised vision backbones. We assessed DINOv3-style pretraining [12] on cell images and declined to run it: the pretraining distribution shares no statistics with sparse calorimeter depositions, and our labelled sample (7×10^4 events) is far below the regime where frozen-backbone transfer beats direct training.

Boosted trees and calorimetry under pileup. Gradient-boosted decision trees [13, 14] remain the standard fast baseline — ATLAS calibrates electron and photon energies with tree-based regression [29] — and ours is specified in Section 3. Learned pileup mitigation began with PUMML [25] and attention-based variants [26]; learned graph-based architectures have been explored for calorimeter energy regression in the CMS context — dynamic graph reduction [27] and deep-learning clustering studies in the ECAL [28]. Those works subtract pileup or regress energy; ours does both in one model and reports where the ceiling comes from. The design of the CMS HGCAL [15, 37] rests on the same premise as our per-bin pattern: fine transverse granularity is there to resolve showers in a dense environment [15], and per-cell timing at the tens-of-picoseconds level is there so that deposits from pileup vertices can be rejected [37] — which is where our own improvements concentrate. The quantile (pinball) loss goes back to [16] and was carried into neural networks by the statistics and forecasting literature [36, 40]; the exponential moving average of weights used here follows the mean-teacher form [38], of which the equal-weight iterate averaging of [17] is the classical ancestor; deep ensembles to [39].

3 Dataset and task

Two simulated samples are used. The *minimum-bias sample* — the physics target and the source of every quoted number — contains 72,554 photon clusters under realistic pileup, spanning the five PicoCal cell-size regions with photon energies restricted to 1–100 GeV. Its region composition (from the cross-validated half: 15 mm ∼12%, 30 mm ∼19%, 40 mm ∼29%, 60 mm ∼34%, 120 mm ∼6%) makes the aggregate metric outer-region dominated, a fact that matters repeatedly below. A *clean sample* of pileup-free photon showers from the same simulation enters only as a training auxiliary (Section 4). Each cluster provides its cells with per-cell energy, position, and two timestamps from the front and back longitudinal compartments — only ∼37% of cells carry a usable front timestamp — plus cluster-level quantities from the standard reconstruction (cluster position, summed energy, cell multiplicity). [Simulation provenance to be completed with the collaboration: generator and simulation framework versions, pileup conditions, out-of-time spillover, timestamp digitisation and resolution, photon selection.] The target is the true photon energy.

Figure of merit. The effective resolution σ_{eff} is the half-width of the minimal interval containing 68.3% of the per-event relative residuals $(E_{\text{pred}} - E_{\text{true}})/E_{\text{true}}$. It is robust to tails (Figure 10 shows why a Gaussian fit would not represent these distributions), and it is bias-blind, which is why we report linearity separately (Figure 9).

We report σ_{eff} in fifteen bins — five regions × three energy terciles, with tercile edges anchored to the starting configuration's energy distribution per region: 46.8/73.8 GeV (15 mm), 31.7/57.5 GeV (30 mm), 19.2/34.0 GeV (40 mm), 12.1/21.5 GeV (60 mm), 9.4/16.5 GeV (120 mm) — plus the event-weighted aggregate, and continuously versus energy in Figure 6. The terciles are computed per region and are therefore not comparable across regions: the median photon energy falls from 59.8 GeV at 15 mm to 12.5 GeV at 120 mm, so the 15 mm "low" bin (20–47 GeV) sits above the entire 120 mm spectrum. Rows of Table 3 should be read down a column, not across one.

Evaluation protocol. The primary protocol is ten-fold cross-validation with a 5% validation slice, so each model trains on 85.5% of events. All ten folds are trained: every event in the sample is predicted exactly once, by an ensemble of five models that never saw it, with the per-event prediction the median over members. The five members per fold are three seeds of the main configuration plus one member with region-restricted recentring and one trained on all cluster cells (Section 4). During development we used a fixed 70/15/15 split with larger seed ensembles; that protocol biases the weak bins optimistically by roughly 0.009 (15 mm low-E: 0.0691 fixed-split vs 0.0783 cross-validated), so we do not quote fixed-split numbers as primary results and we label the protocol wherever one appears. Against selection effects across the 134-configuration campaign (counted as in the introduction, with cross-validated arms counted once; the scored ledger is versioned with the code): the primary protocol was fixed before the final measurement, screening runs never enter the primary claims, and the half-sample preview and its correction (0.0379 → 0.0388) are reported rather than absorbed.

Uncertainties. Two kinds are quoted. On ensemble numbers: a bootstrap error over test events (400 resamples). On claims comparing configurations: the spread of σ_{eff} across independently trained members, which is the appropriate reproducibility scale — a pooled statistical error would assume independent test events that repeat across members. Across the 50 member–fold arms of the primary result, a single member's aggregate σ_{eff} is 0.0397 with spread 0.0010; in the weak bins the single-member spread is 0.0070 (15 mm low-E) and 0.0046 (30 mm low-E).

GBDT reference. The tree reference is one we trained ourselves, on the same events and under the same protocol, rather than a number taken from elsewhere; calling it external would suggest a published result being quoted, and it is not. It is a histogram-based gradient-boosted regressor (300 iterations) on six summary features per cluster — log summed window energy, front/back energy ratio, cell multiplicity, log seed-cell energy, energy-weighted lateral width, and region — trained on the same events with a log-energy target, single model, fixed split. It reaches aggregate $\sigma_{\text{eff}} = 0.1253$ on its own test set (per-region curves in Figure 6). That set is neither the development split nor the cross-validated sample, so the two numbers are not divided directly; instead both estimators are scored on the 10,865 events they share, where the tree reads 0.1189 against the ensemble's 0.0377, a factor 3.15 ± 0.05 with the tree behind in 400 of 400 paired resamples. The comparison holds cell by cell and not only on the aggregate: Table 1 scores both estimators in all fifteen region–energy bins, and the transformer is ahead in every one of them, by between 2.08× and 4.06×. There is no bin in which the tree is competitive, so the aggregate ratio is not an average over a mixed result. It is a fast-baseline reference, not a tuned competitor, and we label it as such.

Table 1: σ_{eff} in all fifteen region–energy bins, both estimators scored on the 10,865 events they share. Terciles are per region and are not comparable across rows. The transformer is ahead in 15 of 15 bins.

	low-E			mid-E			high-E		
region	ours	GBDT	gain	ours	GBDT	gain	ours	GBDT	gain
15 mm	0.0688	0.2793	4.06×	0.0451	0.1368	3.03×	0.0317	0.0938	2.96×
30 mm	0.0669	0.2599	3.88×	0.0392	0.1551	3.96×	0.0281	0.1127	4.01×
40 mm	0.0579	0.1701	2.94×	0.0309	0.0980	3.17×	0.0225	0.0700	3.11×
60 mm	0.0529	0.1504	2.84×	0.0294	0.0822	2.80×	0.0228	0.0646	2.83×
120 mm	0.0635	0.1525	2.40×	0.0379	0.0788	2.08×	0.0269	0.0689	2.56×

One substitution relative to the original project plan is made explicit: the LHCb production clustering algorithms (the Cellular Automaton and its graph-clustering successor) run upstream of this dataset — the clusters we receive *are* their output — so their energy estimate enters our comparisons as the calibrated cluster sum ($\sigma_{\text{eff}} = 0.184$ on this sample, Figure 16) rather than as re-implementations.

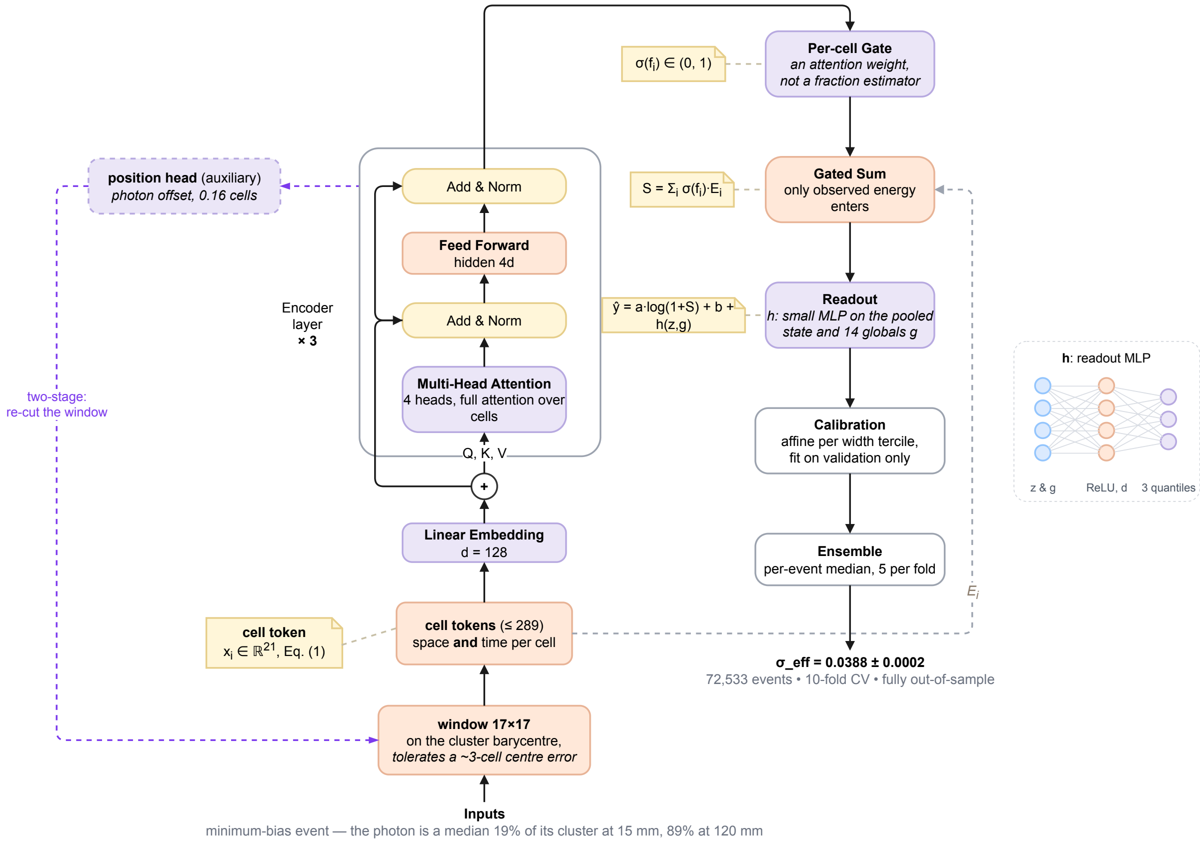

Figure 2: The SpaceTformer pipeline. Each cell is a token in detector space and time; the encoder is a plain transformer over cell tokens only; the cluster-level global features g enter only at the readout, which is a gated energy sum in log space plus a small correction h; the loss is a three-quantile pinball loss with EMA weights; predictions are the per-event median over ensemble members; the inset draws h itself, a one-hidden-layer network from the pooled state and the globals to the three quantiles. The decisive components are the window's size and centring at the bottom (Section 5.2). Dashed purple: the auxiliary position head and the two-stage path that re-cuts the window around the predicted photon position (Section 5.3).

4 Methods

Figure 2 shows the full SpaceTformer pipeline. Throughout, the *seed cell* is the highest-energy cell used by the standard reconstruction to anchor a cluster, and *ring* r denotes the square shell of cells at Chebyshev distance r from the window centre.

Tokens. Each fired calorimeter cell i of the window becomes one token. Sixteen channels are the detector reading itself,

$$\mathbf{x}_i^0 = \Big(\log(1+E_i),\ \log(1+E_i^{\mathrm{f}}),\ \log(1+E_i^{\mathrm{b}}),\ \delta_i^u,\ \delta_i^v,\ r_i,\ \log p,\ \Delta t_i^{\mathrm{f}},\ \Delta t_i^{\mathrm{b}},\ m_i^{\mathrm{f}},\ m_i^{\mathrm{b}},\ \mathbf{1}_\rho\Big) \in \mathbb{R}^{16}, \tag{1}$$

where E_i is the cell energy and $E_i^{\mathrm{f}}, E_i^{\mathrm{b}}$ its front and back longitudinal samples; (δ_i^u, δ_i^v) is the offset from the window centre in cells and r_i its Chebyshev modulus; p is the cell pitch; $\Delta t_i^{\mathrm{f,b}} = t_i^{\mathrm{f,b}} - \mathrm{med}_j\, t_j^{\mathrm{f,b}}$ are the two timestamps with the window median removed and clipped to $\pm 5\,\mathrm{ns}$; $m_i^{\mathrm{f}}, m_i^{\mathrm{b}} \in \{0,1\}$ record whether each timestamp exists; and $\mathbf{1}_\rho$ is a five-way one-hot of the calorimeter region. The two time channels are arrival times, not an engineered pull: subtracting the window median removes only the event's overall time offset. A cell with no

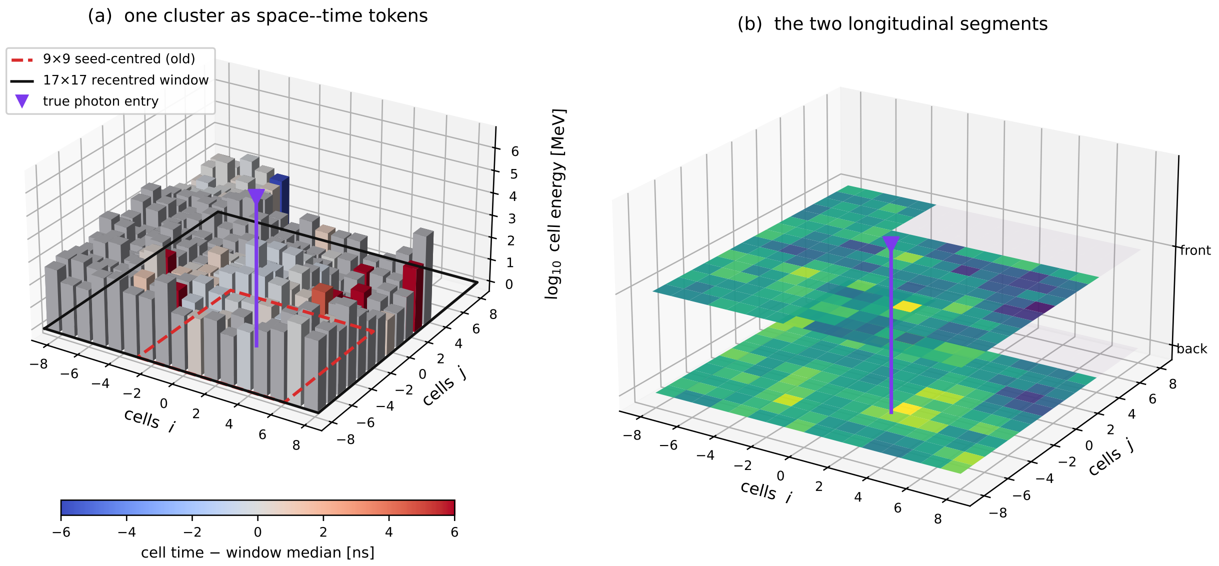

Figure 3: One minimum-bias cluster as the model receives it (15 mm region, $E_\gamma = 37\,\mathrm{GeV}$, 424 fired cells in the cluster, of which the window admits at most 289). (a) Each fired cell is one token: its position in the 17×17 window, its energy (bar height) and its timestamp relative to the window median (colour; grey bars carry no timestamp). Pileup deposits fill the window at all times, while the photon's own cells arrive together near zero. (b) The same cluster in its two longitudinal segments, the depth information that separates the start of the shower from its tail.

usable timestamp carries $m = 0$ beside a zero, which after the linear embedding is equivalent to a learnable missing-value embedding.

Every member of the primary result carries five further channels, added because the ablation of Section 5.9 measured them worth $0.0409 \to 0.0389$ together: the cell's offset from the reconstructed cluster centroid in units of the pitch, $(x_i - x_c)/p$ and $(y_i - y_c)/p$, and the same three energies again normalised for the cell area, $\log(1+E_i) - 2\log(p/15\,\mathrm{mm})$ and its two longitudinal counterparts, so that a deposit means the same thing in a 15 mm cell as in a 120 mm one. The token the encoder actually embeds is therefore $\mathbf{x}_i \in \mathbb{R}^{21}$, and a reader reproducing Table 3 needs all twenty-one. The whole set $\{\mathbf{x}_i\}$ is what the encoder attends over; nothing is aggregated before it. Cluster-level reconstruction quantities enter as global features. There are fourteen: five describing the window itself ($\log(1+\sum_i E_i)$, the seed energy, $\log n$ for n fired cells, the front/back energy ratio, and the energy-weighted lateral width), eight relating the window to the cluster the standard reconstruction found (its total energy, the fraction of it the window captures, the same fraction front and back separately, the cluster's own front/back ratio, the number of reconstructed clusters in the event, and the two components of the window centre's offset from the cluster centroid), and one indicator marking events drawn from the clean auxiliary sample rather than from minimum bias. All inputs come from the standard reconstruction or from that sample label; none uses generator truth. Figures 3 and 4 show the same real input from two views: the token set as the detector produces it — one box per fired cell, in space, in time, and split over the two longitudinal segments — and the same event in projection, where the pileup blanket the model must subtract, the sparsity of the timestamps, and the three candidate window centres of Section 5.2 are read off directly.

Window. The decisive design choices are *where* the window sits and *how much* it covers, not what reads it. The main configuration uses a 17×17 window — half-width $w = 8$ cells, so $(2w+1)^2 = 289$ grid positions — centred on the cluster energy barycentre (computable from

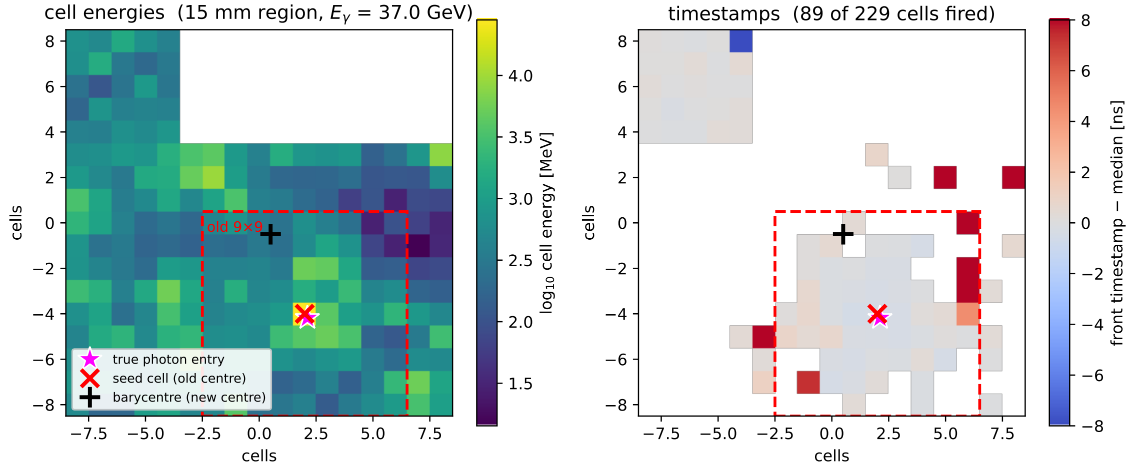

Figure 4: One real minimum-bias input to SpaceTformer: a 15 mm-region cluster ($E_\gamma = 37$ GeV, 424 fired cells in all, of which the 17×17 window admits at most 289) shown in the recentred window. Left: per-cell energies — pileup deposits everywhere, with 74% of the cluster energy outside the starting 9×9 window (red dashed). Right: front timestamps, carried by only 39% of the fired cells. Markers: true photon entry (star), seed cell (cross, the old window centre), energy barycentre (plus, the new centre).

standard reconstruction quantities, hence available online) in *all* regions; one ensemble member instead restricts the recentring to the two inner regions, where the measured mis-seeding rate is highest and the gain concentrates. Section 5.2 presents the measurements behind both choices.

Encoder. SpaceTformer's encoder is a plain token transformer [3]: $d = 128$, 4 heads, 3 layers, feed-forward width $4d$, dropout 0.1, full self-attention over the cell tokens. The fourteen cluster-level global features do not enter the encoder; they are concatenated to the pooled state at the readout. Attention therefore operates on per-cell channels, with the caveat that two of them — the cell pitch and the region one-hot — are constant within an event and tell the encoder which region it is reading. The name marks what the model attends over — each token is a point in detector space *and* time, with the timestamps worth 20% of the aggregate resolution (Section 5.8) — not a factorised attention scheme: we measured time-factorised attention and it loses. Ten encoder families were tried against this one (Section 5.6); none won.

Physics-constrained readout. Instead of predicting the energy freely, the network emits a per-cell gate $\sigma(f_i) \in (0, 1)$ and the readout is

$$\hat{y} = a \cdot \log\Big(1 + \sum_i \sigma(f_i)\, E_i\Big) + b + h(z, g), \tag{2}$$

a gated energy sum in log space plus a small learned correction h (an MLP on the pooled encoder state z and the global features g). The gated sum is the dominant pathway and anchors the prediction to observed cell energies; the correction term is free in log space and absorbs what the sum cannot express, so the constraint is an inductive bias, not a hard bound. This stabilises the low-statistics bins. We stress, and measure in Section 5.7, that the learned gate is *not* a per-cell pileup-fraction estimator and forcing it to be one hurts.

Loss and target. The choice was measured, not assumed (Table 2, all arms sharing this encoder). Predicting the energy directly and predicting a residual on the calibrated sum are indistinguishable and poor; the gated readout with a Huber loss halves the error; quantile losses beat Huber. The production loss is a pinball loss at quantiles $(0.25, 0.5, 0.75)$ — the median is the prediction — plus a width-regularisation term that penalises wide inter-quantile intervals while softly enforcing their coverage. One arm deserves its warning label: *trimmed risk* — dropping the worst 10% or 20% of events from the loss, a textbook move for a robust target metric — is catastrophic (0.0513 and 0.0565) and unstable across seeds (± 0.05 at 15 mm low-E). The events it drops are precisely the pileup-heavy population of Section 5.4; refusing to train on them removes the hard half of the physics.

Table 2: Loss and target study, identical encoder, fixed-split protocol, aggregate σ_{eff} (five seeds unless noted).

target / loss	note	aggregate
direct energy, L2		0.0650
residual on calibrated sum, L2		0.0675
gated readout, Huber	first structural gain	0.0462
gated readout, 3-quantile		0.0427
+ EMA + clean-sample auxiliary	frozen recipe	**0.0390**
trimmed risk, drop worst 10%	2 seeds	0.0513
trimmed risk, drop worst 20%	2 seeds	0.0565

Clean-sample auxiliary. Training batches mix the minimum-bias events with events from the pileup-free clean sample (Section 3), sharing the same loss and readout. The clean events anchor the photon response — what a shower looks like when nothing contaminates it — while the minimum-bias events teach the subtraction; removing the auxiliary costs resolution and it is part of the frozen recipe (the `CleanAux` tag in every ledger name).

Training and calibration. AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}), batch 96, cosine-annealed learning rate, up to 100 epochs with early stopping on the validation loss (patience 15), and an exponential moving average of the weights (decay 0.999, the mean-teacher form [38]) used for evaluation. A six-arm sweep of this recipe at the end of the project (learning rate up and down, batch up and down, coordinate jitter at two amplitudes) returned 0.0422–0.0462 against the frozen recipe's 0.0402 on the same split — the recipe was already at its optimum. A post-hoc code audit found two schedule defects, disclosed for reproducibility: later runs inadvertently stepped the cosine schedule twice per epoch (halving its effective period), and a seventh sweep arm intended to toggle the schedule was inert and is excluded above; every comparison in this paper is between runs sharing the same code era, so no paired conclusion is affected. Post-hoc calibration is a per-group affine correction with events grouped into terciles of the predicted inter-quantile width (a per-event uncertainty proxy); under cross-validation it is fitted per fold on that fold's own validation slice, never on pooled or test events.

Ensembling. Per-event median over five members per fold: three seeds of the main recentred configuration, one member recentring only the two inner regions, and one member trained on all cluster cells rather than the fixed window. The median interacts correctly with the robust σ_{eff}; a learned stacking combination was measured and is worse (Section 5.9).

5 Results

5.1 Primary result: out-of-sample resolution

Table 3 is the main results table; Figure 6 shows the same information continuously in energy against the GBDT reference, and Figure 9 shows the response linearity that σ_{eff} cannot see. All numbers are cross-validated and out-of-sample over all ten folds, five members per fold.

Table 3: Primary result: σ_{eff} per region and energy tercile, ten-fold cross-validation, all folds complete, out-of-sample (72,533 events). Parentheses give the bootstrap error on the last digits; n is the number of test events per bin (low/mid/high). "Start" is this project's starting configuration (fixed 9×9 seed-centred window, five-seed fixed-split ensemble), shown for the two regions where it failed the per-bin target; its aggregate was 0.0390. The sample holds 72,554 clusters; the per-fold ensemble merge keys events by (energy, region) and 21 events (0.03%) collapse under identical keys.

region	low-E	mid-E	high-E	n (l/m/h)	start, low-E
15 mm	0.0783(22)	0.0451(11)	0.0340(8)	2815/2866/2685	0.1654
30 mm	0.0778(19)	0.0422(7)	0.0293(5)	4458/4828/4693	0.0957
40 mm	0.0578(10)	0.0316(5)	0.0240(4)	6977/6908/6982	
60 mm	0.0529(7)	0.0310(3)	0.0232(3)	8200/8478/8259	
120 mm	0.0642(22)	0.0367(11)	0.0291(8)	1546/1528/1310	
aggregate		0.0388 ± 0.0002		72533	0.0390

Every bin is below 0.079; thirteen of fifteen are below 0.065. The two weakest bins are the low-energy bins of the two innermost regions, where pileup contamination is worst and event counts smallest. Figure 5 shows all fifteen bins against the starting configuration: the gains concentrate exactly where the containment measurement (Section 5.2) says they should — the inner-region low-energy bins ($0.165 \to 0.078$ and $0.096 \to 0.078$) — with small regressions in the outer bins that were already contained, the largest at 120 mm low-E ($0.0529 \to 0.0642$; the starting value coincides numerically with the 60 mm low-E entry above it). At 120 mm the barycentre is dragged by pileup tails while the seed cell was already correct, so recentring is measured to cost resolution there — yet four of the five ensemble members use it; the fifth, region-restricted member limits the damage.

Why the curve stops falling at the top. A resolution-versus-energy curve on this sample is not one detector measured at several energies: it is a different mixture of regions at each point. Below 10 GeV the sample is 60% 60 mm and holds no 15 mm clusters at all; above 80 GeV it is 58% 15 mm and 42% 30 mm, with nothing from the outer regions. Energy and pseudorapidity are tied together at fixed transverse energy, so climbing the axis walks inwards across the calorimeter and into the worst of the pileup. The flattening above about 20 GeV, and the point-to-point wobble past 80 GeV — 2.49% at 80–90 GeV against 2.80% at 90–100 GeV, a 1.5σ difference on statistical errors of 0.13 and 0.19% — are that composition changing under the curve, not the estimator degrading. Read the per-region curves for detector behaviour and the aggregate curve for what a mixed sample delivers.

Figure 8 fits the standard calorimeter parameterisation $\sigma_{\text{eff}}(E) = a/\sqrt{E} \oplus b/E \oplus c$ to the cross-validated points per region. The fit summarises the physics compactly: the $1/E$ term — which for a pileup-dominated sample plays the role of the noise term, an energy-independent contamination in GeV — is largest by far in the 15 mm region ($b = 2.1$ GeV against 0.16–0.34 GeV in the outer regions), while the constant terms sit below 2.5% everywhere. The residual low-energy weakness of the inner regions is pileup energy, not sampling statistics. The fit quality has to be stated with the parameters: χ^2/ndf is 1.0 at 120 mm, 1.6 at 60 mm, 2.3 at 40 mm and 3.9 at

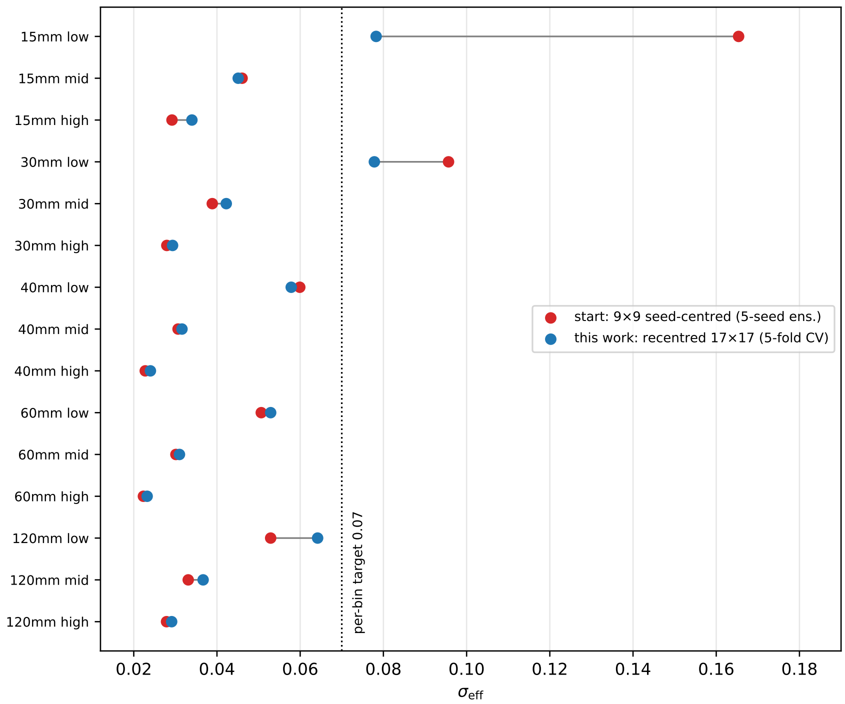

Figure 5: All fifteen region–energy bins: starting configuration (red, fixed-split five-seed ensemble) versus this work (blue, cross-validated ensemble). The two protocols differ slightly (Section 3); the dominant moves — 15 mm low 0.165 → 0.078, 30 mm low 0.096 → 0.078 — dwarf that difference.

30 mm, but 23 at 15 mm, where the stochastic and constant terms both run onto their bounds. The three-term form is a poor description of the innermost region, and the $b = 2.1$ GeV there should be read as a statement that the $1/E$ term dominates, not as a measured noise constant.

Against the calorimeter LHCb has now. The natural external scale is the existing LHCb electromagnetic calorimeter, whose simulated resolution runs from about 11% at low energy to about 4% at very high energy, integrated over its areas [19]. Binned the same way, this work reads 8.0% below 5 GeV, 4.1% at 10–20 GeV, 3.0% at 30–50 GeV and 2.6% above 80 GeV, on a minimum-bias sample — that is, *with* the pileup, which the reference curve does not carry.

Three things make this a scale rather than a like-for-like comparison, and we state them rather than let the numbers stand alone. The detectors differ: PicoCal is the Upgrade II replacement, with finer granularity and per-cell timing. The statistics differ: the reference is a Gaussian width, ours the half-width of the 68.3% interval, which is the more conservative of the two on a distribution with tails. And the tasks differ: the reference measures a calibration chain, ours a regression trained on the same simulation it is scored against. What the comparison does establish is that the numbers here sit at the scale of a working calorimeter's performance rather than beside it.

5.2 Origin of the improvement: window containment and centring

The improvement over the starting configuration came from two reconstruction-level corrections, and this subsection presents the measurements that found them, in the order they happened. Figure 11 first shows why the difficulty is not shared equally across the calorimeter.

The window half-width had been fixed at w=4 (9×9 cells) from the start of the project and never varied. Figure 12 shows the measurement that broke this: the mean contained fraction of

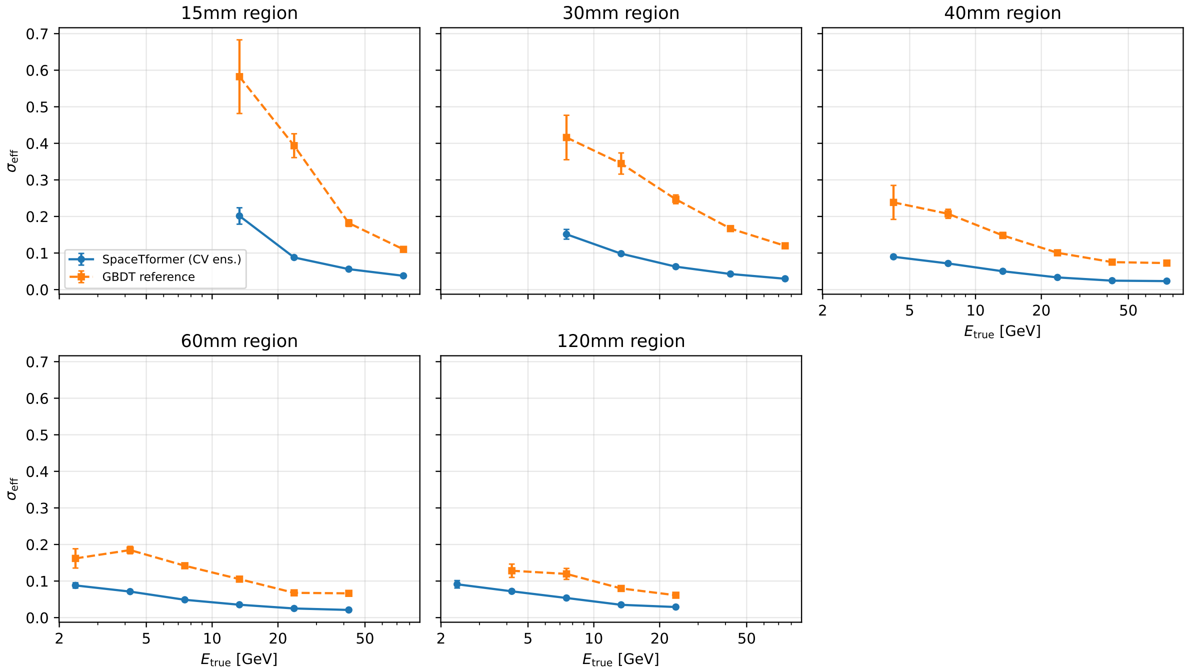

Figure 6: σ_{eff} versus true photon energy per region: SpaceTformer (cross-validated ensemble, circles) against the six-feature GBDT reference (single model, fixed split, squares). Error bars are bootstrap errors over test events. The gain is largest at low energy and in the inner regions, matching the pileup-contamination pattern.

cluster energy versus window half-width, per region. At w=4 the seed-centred window holds 41% of the 15 mm cluster's energy against $\geq$ 92% in the three outer regions — one shared window width had been silently tuned for the abundant outer regions.

Reading that figure as photon containment would be a mistake, and separating the two is what explains the size of the gain. The cluster is photon plus pileup. The *photon* on its own is compact: measured on the pileup-free clean sample with the window on the seed cell, its median contained fraction is 94.4% at w=1 and 99.8% at w=4 in the 15 mm region, and at least that in every other region. A seed-centred 9×9 window would already hold all of the signal.

The window is not seed-centred, though: it is placed on the reconstructed barycentre, which pileup drags a median 3.1 cells away from the photon at 15 mm. Measured on the paired sample, where the photon's contribution to each cell is known, a *barycentre*-centred window at 15 mm holds a median 1.2% of the photon at w=1, 56% at w=2, 97% at w=3 and 99.6% at w=4 — and the tail matters more than the median: the fraction of events keeping less than 90% of the photon is 73%, 38%, 7.8% and 0% at w=2, 3, 4, 6. The wide window is therefore not compensating for a wide shower. It is making the aperture robust to an imprecise centre, and the region where the centre is worst in cell units is exactly the region the window fixed.

Above w=6 the photon is fully contained in every event, yet the window scan below keeps improving to w=8–10. That remaining part is not coverage: the extra cells carry no signal, and what they carry is the pileup field the estimate has to subtract — at 15 mm the 17×17 window holds 3.6 times the photon's own energy, over a median of 160 fired cells (Figure 11). Two controls separate the roles. For a plain calibrated sum with no model, widening the window makes the 15 mm low-energy bin monotonically *worse* (0.36 → 0.53 from w=1 to w=8), so the outer cells are not signal to be added. And restricting the network's gated sum to a central core while the encoder still attends over the whole window costs 15 mm low-E 0.0863 at core w=4 and 0.1270 at core w=2 against 0.0755 for the full sum — tracking the truncation tail above almost exactly. That the damage is truncation and not the core idea is confirmed by repeating

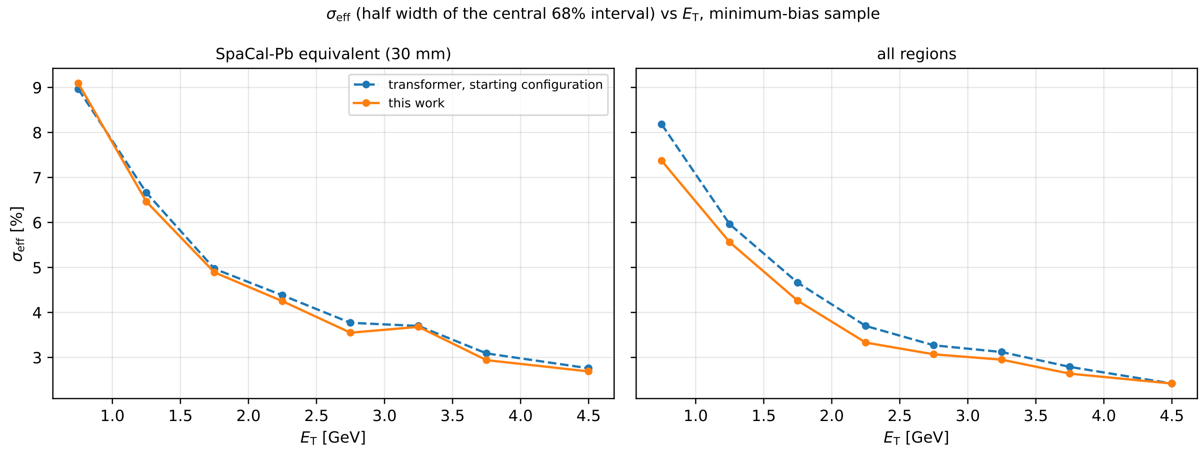

Figure 7: The same resolution against *transverse* energy, which is the axis the graph-network work on this detector reports [18] and the one in which the physics selections are written. Left: the 30 mm region, which is the SpaCal-Pb equivalent — their shashlik modules are 4×4 cm, our 40 mm, and SpaCal-Pb sits between the tungsten spaghetti calorimeter and the shashlik. Right: all regions. Dashed is the starting configuration, solid this work. Plotted this way the two studies can be laid side by side: the metric is the same σ_{eff}, and this sample already lies inside their 0.5–5 GeV range. It is an axis, not a claim about their numbers — their photons are isolated with pileup overlaid rather than real minimum-bias events, and their pileup condition is the worst case rather than the nominal one.

it on a window centred by the learned pointer of Section 5.3, whose median error is 0.16 cells: there a w=2 core scores 0.0389, equal to the same configuration with the full sum, where on a barycentre-centred window it scores 0.0436. With an accurate centre the energy sum needs 25 of the 289 cells and the rest are read but never summed — no resolution is gained, but the readout becomes a much smaller object than the window it sits in. The window plays both roles, and the measurements say where each one takes over.

Sweeping the seed-centred window from 5×5 to 21×21 (Figure 13) confirms the diagnosis in resolution space: 15 mm low-E falls from 0.186 to 0.102 as the window opens — the single largest lever found in the whole project — while the *aggregate barely moves* (0.038–0.041 across the whole scan), because the outer regions dominate the event count and were already contained. That flatness is exactly how the defect stayed invisible: the aggregate metric hid it. Ring summaries (coarse sums of the cells outside the window, a spatial analogue of the token patching of [6]) let w=8 match w=10 at half the tokens, but only extend a resolved core; at w=4 they recover little of what truncation destroys.

The second correction is *where* the window sits. Re-centring it on the cluster energy barycentre gives the biggest single step in the cumulative decomposition (Table 4: 0.1156 → 0.0711), and it invites a wrong reading — that the barycentre localises the photon better than the seed cell. That reading is measurably false. One term needs pinning down before the comparison, because it is easy to read the wrong way. The recentred window is placed on the centre of the seed *module* — the 12×12 cm block that holds between one and 64 cells — not on the centre of the seed *cell*. In the 15 mm region a module is eight cells across, so the two differ by up to four cells, and it is the module centre, not the cell, that the measurements below are about.

That choice is a trade the paper should state rather than sell. Anchoring the aperture to the module buys coverage at low energy in the inner regions, where the pileup-dragged barycentre is the problem; it costs position resolution, because the frame the network sees is quantised to the module and no longer follows the photon. This work measures energy, so the trade is worth taking here; an analysis that needs the photon's position would want the opposite.

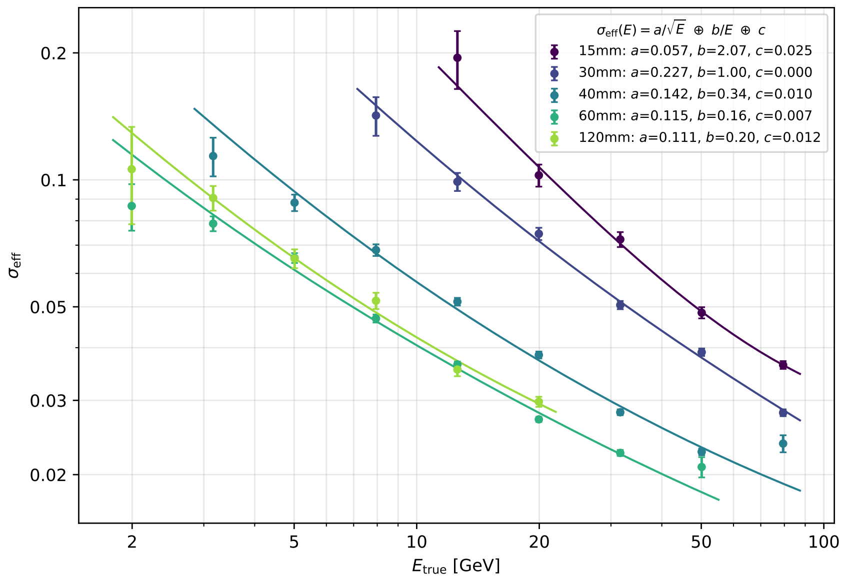

Figure 8: σ_{eff} versus true energy per region (cross-validated ensemble, bootstrap errors) with the standard parameterisation $\sigma_{\mathrm{eff}}(E) = a/\sqrt{E} \oplus b/E \oplus c$ fitted per region. The b (noise) term absorbs the energy-independent pileup contamination and is an order of magnitude larger in the 15 mm region than in the outer ones.

Figure 14 shows the distance of each candidate centre from the true photon entry point in the 15 mm region: the seed cell is *much closer* to the photon (median 0.44 cells against 3.06), yet the barycentre-centred window performs better. The window is not a photon-pointer; it is an aperture, and what matters (Figure 12, solid vs dashed) is that the aperture covers the energy field — photon tail plus the pileup the model must subtract — and gives the network a coordinate frame anchored to that field rather than to one volatile cell. (The decomposition in Table 4 ends at the cross-validated 0.0783.) Centre estimators aimed at better photon localisation (an argmax-iterated log-centroid, a hybrid that follows the seed except on disagreement, a clustered 3×3 seed) were all measured worse than the plain barycentre.

Table 4: Cumulative decomposition of σ_{eff} in the 15 mm low-energy bin. The first three rows use the development protocol (fixed split, five-seed ensemble); the last row is the out-of-sample cross-validated number, higher because its per-fold ensembles are smaller — the two protocols are not directly comparable and both are shown deliberately.

step	15 mm low-E	protocol	mechanism
9×9 seed-centred window	0.1654	fixed split	centre error truncates signal
widen to 17×17	0.1156	fixed split	robust aperture + pileup context
+ re-centre on barycentre	0.0711	fixed split	smaller centre error (see text)
cross-validated ensemble	0.0783(22)	CV	primary protocol

5.3 A learned two-stage window

The centring analysis leaves an open question: the barycentre-centred window gains through error-tolerant coverage while the seed is the better locator (Figure 14) — what would a window centred on the *true* photon entry add? Retraining the final configuration with the window centred on the true entry point (a diagnostic bound: truth is not available at inference) gives a single-member aggregate of 0.0383 against the five-seed member mean 0.0399 ± 0.0005, with the

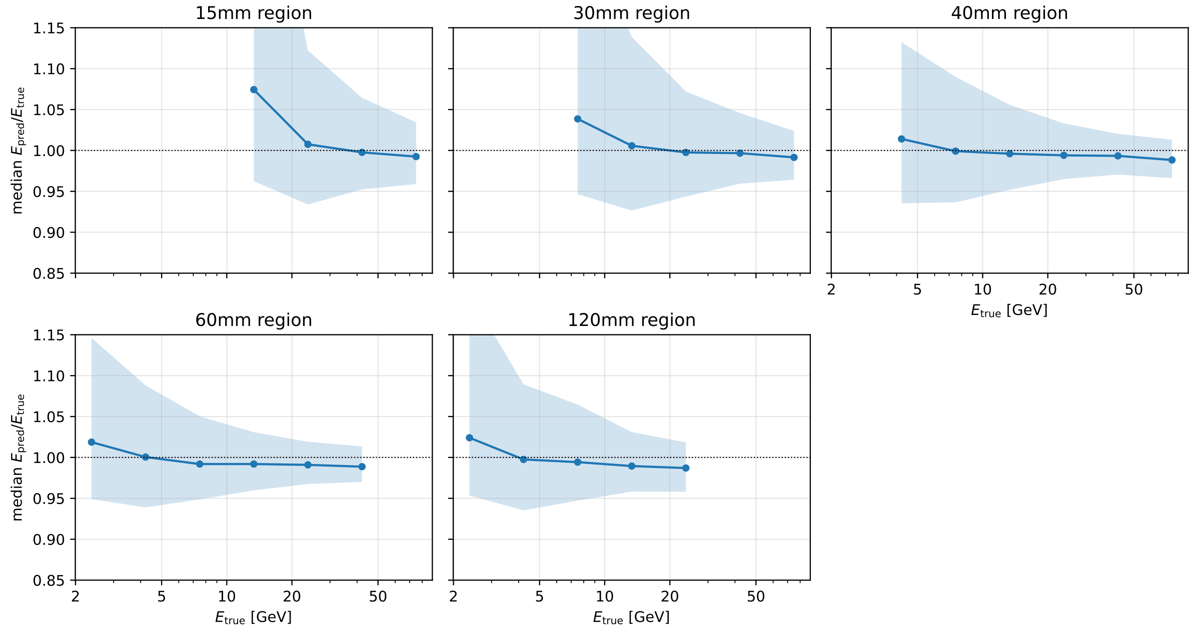

Figure 9: Response linearity: median $E_{\mathrm{pred}}/E_{\mathrm{true}}$ versus true energy per region for the cross-validated ensemble, with the 16–84% band shaded. σ_{eff} is bias-blind, so this plot carries the scale information the resolution table cannot.

weak bins improving by ∼10% — perfect centring is worth 4% aggregate, and it is the largest in-data margin any single mechanism measurement left on the table.

That bound is reachable because the network can point. The auxiliary position head (a three-unit regression of the photon's seed-relative offset and time, trained alongside the quantiles at weight 0.2) turns out to localise the photon with a median error of 0.164 cells and a 90th percentile of 0.389 cells over the full sample — against the seed's 0.43-cell median with its 8.5-cell 90th percentile tail at 15 mm, and the barycentre's 3.1-cell median drag. On its own, adding this head already helps the energy: ten retrained members average 0.0395 ± 0.0004 and their ensemble reaches 0.0380 (paired bootstrap against the reference ensemble: -0.0008 ± 0.0003, better in 400/400 resamples; the first five members alone give 0.0382).

The two-stage architecture closes the loop: a first pass points at the photon, the window is re-cut around the pointed position, and a second pass regresses the energy from the re-pointed window — the learn-where-to-look-then-measure pattern of spatial transformer networks [30] and deformable attention [31], applied to a domain where every production reconstruction uses a fixed seed- or barycentre-centred window. Measured offline (pointer predictions from the trained auxiliary members; the test split is pointed by models that never saw it), the five retrained members land at 0.0389 ± 0.0003 — the five best single members of the 134-configuration campaign.

Table 5 gives the ensembles. The gain concentrates exactly where the centring analysis said it would: 15 mm low-E 0.0670 against the reference ensemble's 0.0711 and 30 mm low-E 0.0675 against 0.0717 (−6% each). The true-entry configuration is not a like-for-like comparison at bin level — it was measured with a single member, not an ensemble — so it bounds the aggregate gain rather than the per-bin one. The one regression is the outermost region (120 mm low-E 0.0581 vs 0.0555), consistent with the earlier region-restricted recentring measurement: where occupancy is low the seed already is the photon, and re-pointing only adds noise — the pointer should be applied in the inner regions only. Distilling the five-seed ensemble into a single member is a second route to the same margin (0.0383 ensemble, P=0.98; a distilled single member matches the ensemble's accuracy class at one fifth of its inference cost, which matters

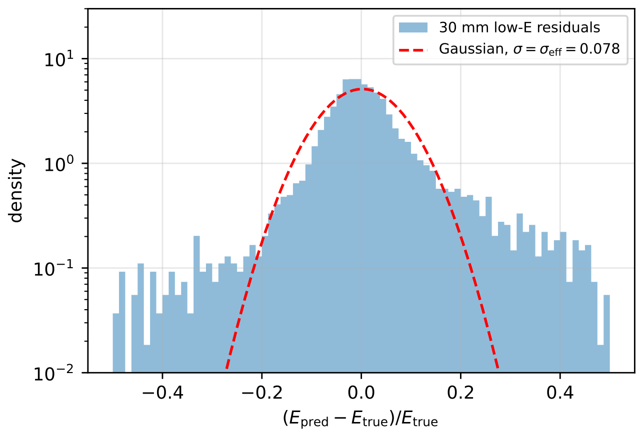

Figure 10: Relative residual distribution in the 30 mm low-energy bin (cross-validated ensemble, log scale) against a Gaussian of width σ_{eff}. The non-Gaussian tails are why σ_{eff} is the figure of merit and why tail-reshaping interventions do not move it.

Table 5: The pointing ladder. All rows use the development protocol — the fixed 70/15/15 split, seed-ensembled by per-event median — and paired bootstrap over 400 shared resamples against the reference ensemble. The reference row happens to read 0.0388 here as well, numerically equal to the cross-validated primary result of Section 5.1 by coincidence: the two are computed on different event sets under different protocols and must not be compared with each other. Only differences *within* this table are meaningful.

ensemble	members	aggregate σ_{eff}	paired Δ vs reference
reference (recentred window)	5	0.0388	—
+ position head	10	0.0380	-0.0008 ± 0.0003, P=1.00
two-stage (point, re-cut, measure)	5	0.0378	-0.0010 ± 0.0003, P=1.00
pooled across the three families	20	0.0376	-0.0013 ± 0.0002, P=1.00

for the trigger budget of Section 5.10). The two routes do not stack (-0.0002 ± 0.0003, P=0.65 between them): both recover the same centring-and-averaging margin, one structurally and one by imitation.

The gain does not survive cross-validation. Every value above is a development-protocol number, and we ran the obvious check: repeat the construction inside the ten-fold protocol, training the position head on each fold and building that fold's pointer from it, so nothing about the photon position leaks into the fold's test events. Three folds were completed before the run was stopped. The pointer is unaffected — its median error is 0.174 cells per fold against 0.164 on the development split — but the energy gain is not there. Scored on identical events, the two-stage member reads 0.0388 and 0.0387 on folds 0 and 1 against the auxiliary-head member's 0.0383 and 0.0385; pooling the three cross-validated families reaches 0.0378 where the primary recipe reads 0.0381 on the same events, a 0.8% difference rather than the 2.6% the development split showed.

We report this rather than quietly keeping the favourable protocol. The development-split measurement is correct as made — five members, a paired bootstrap better in 400/400 resamples — and it does not transfer. The most likely reading is that the fixed split's test set is both smaller and easier than the full sample, so a mechanism worth a genuine but small margin can look larger there than it is. The auxiliary position head, by contrast, is worth about 1% under both protocols, which is the level at which we would now describe both mechanisms. The centring

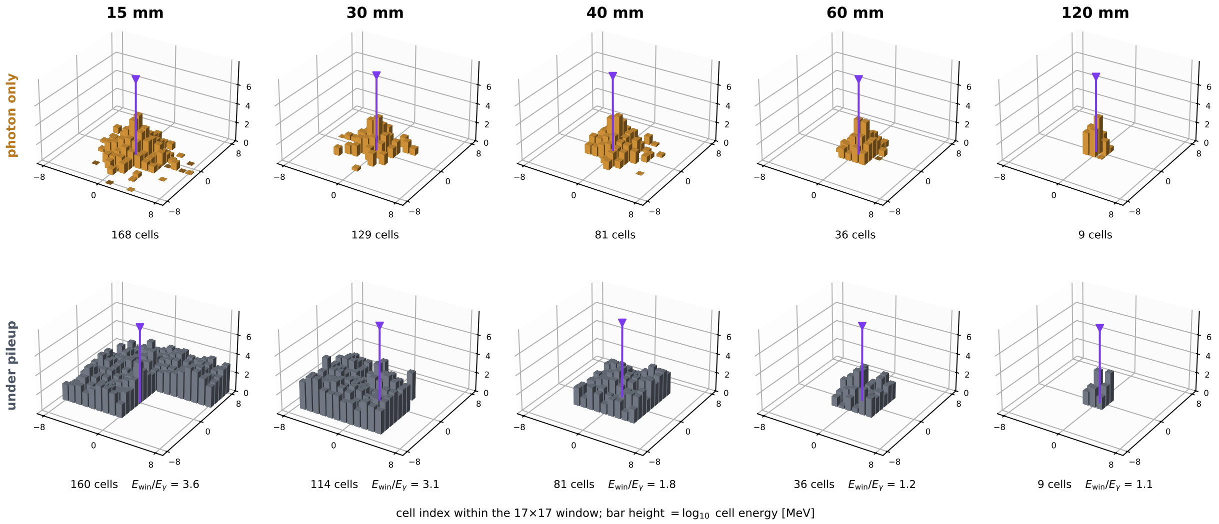

Figure 11: Why the inner regions are the hard ones. Each column is a cell pitch; each panel shows one low-energy event of that region in the 17×17 window, with the true photon entry marked. Top: the pileup-free clean sample, so the bars are the photon alone. Bottom: the minimum-bias sample, the same region under pileup. Annotations are medians over the whole sample, not properties of the displayed event. The photon itself is compact everywhere — 94–100% of its energy falls within 3×3 cells of the seed on the clean sample — so what changes across regions is the *load*: the window collects 3.6× the photon's energy at 15 mm and 1.1× at 120 mm, spread over 160 against 9 fired cells. At 15 and 30 mm the estimator therefore has to subtract about three quarters of what it is given, and at low photon energy that subtraction is the whole measurement.

bound of 4% stands; what is uncertain is how much of it any estimator can actually take.

5.4 Decomposition of the remaining error

With the window fixed, we dissected what is left, joining per-event truth diagnostics to the best ensemble's residuals (5,372 joined events; diagnostics are truth-based and used only for analysis, never as inputs). Four measurements structure the remainder.

(1) The core width is pileup magnitude, not a missing mechanism. Splitting each weak bin at the median of the ratio of window energy to true energy — a direct measure of how much non-photon energy the window holds — separates two populations (Table 6). Every low-energy bin tested, including 120 mm which was never window-limited or mis-seeded, is a mixture of a clean population already at 0.04–0.07 and a pileup-heavy population near 0.10. The inputs that describe pileup magnitude (window-to-cluster energy ratio, occupancy, density) are already features, so the model knows which population an event belongs to and still cannot delete the fluctuation: this is the aleatoric component, localised. It is what motivates reading the residual as a statistics problem (Section 5.11) rather than a modelling one.

Table 6: σ_{eff} of each low-energy bin split at the median per-event pileup contamination (truth-based diagnostic). The pileup-heavy half sits near 0.10 in every region — including 120 mm, which never had a window or centring problem.

bin	cleaner half	pileup-heavy half
15 mm low-E	0.0683	0.0895
30 mm low-E	0.0598	0.0977
120 mm low-E	0.0370	0.0977

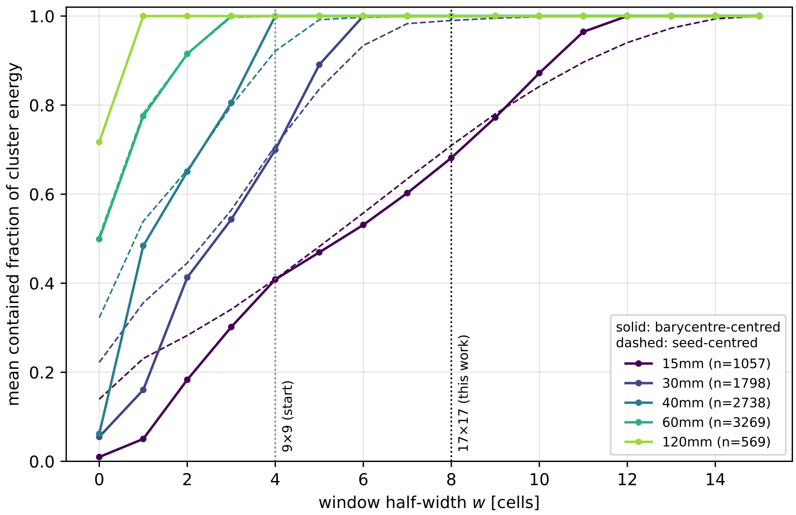

Figure 12: Mean contained fraction of cluster energy versus window half-width w, per region, on the minimum-bias sample (12 files, 9,431 events), for barycentre-centred (solid) and seed-centred (dashed) windows. The vertical lines mark the starting 9×9 window and this work's 17×17. At w=4 the 15 mm region is the only one below half containment.

(2) The tails are catastrophic and separate from the core. The ratio of the 95% to the 68% residual quantile is ≈ 5 in all three bins of Table 6, against 2.0 for a Gaussian: a 5–10% population with $|\text{res}| \sim 0.4$–0.6 sits far outside the core (Figure 10). σ_{eff} barely sees these events, so no tail-directed intervention moves the aggregate number — but they matter for physics use, and their likely origin (wrong-photon matches, windows dominated by an overlapping cluster) is a reconstruction question upstream of regression. For deployment, the model already carries a per-event confidence: the predicted inter-quantile width used by the calibration (Section 4) widens on exactly these events and can flag them at inference without any additional head.

(3) Ambiguity is measured, priced in, and still costs. After recentring, 30 mm low-E events where the barycentre and the seed cell agree within one cell resolve at 0.0489; the rest at 0.0796. The disagreement magnitude is observable at inference and is already an input — the model knows which events are ambiguous and the gap persists, which again says magnitude of ambiguity, not missing information.

(4) Timing availability correlates the right way. Events with more time-stamped cells do better (15 mm low-E: 0.0706 with above-median timing coverage vs 0.0823 below), consistent with the timing ablation (Section 5.8) — denser timing coverage in the detector design would feed this model directly.

5.5 What limits the measurement

The four measurements above say what the residual is made of. This subsection prices it: how much of the aggregate each part carries, how much of each part any estimator could remove, and what is therefore left. All values here use the development protocol on the pooled 28-member ensemble (0.0376 aggregate, 10,877 events), because it is the largest set of members we have; the conclusions are ratios and carry over.

Where the aggregate comes from. Table 7 gives, per region–energy bin, its share of the sample and what the aggregate would become if that bin were predicted perfectly. The bins with the worst resolution are not the ones that matter most: 15 mm low-E is 3.8% of the sample and worth 0.0022 on the aggregate, while 60 mm low-E is 11.3% and worth 0.0057. Every window,

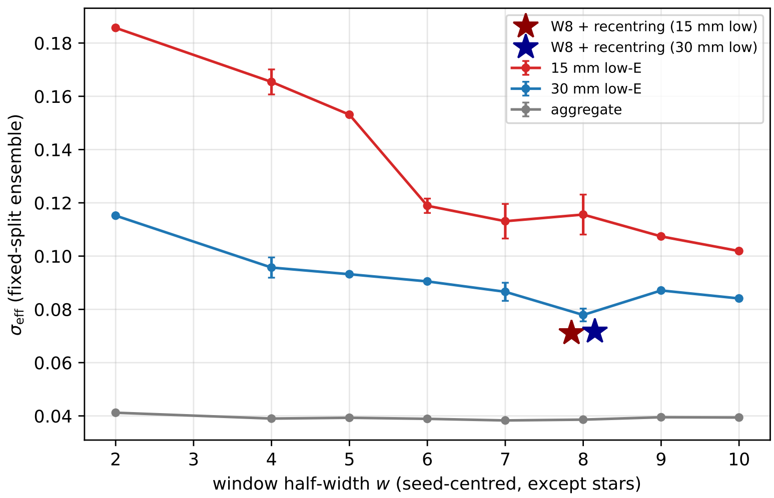

Figure 13: The window scan: σ_{eff} versus window half-width, seed-centred (circles; fixed-split ensembles, error bars where multi-seed), with the recentred w=8 configuration as stars. The aggregate is flat while the worst bin halves: the aggregate metric hid the defect.

centring and pointing measurement in this paper was developed against the inner regions; the aggregate is carried by the middle ones.

Table 7: Error budget by bin, development protocol. "Aggregate if perfect" replaces that bin's residuals with zero and rescores everything; it is an upper bound on what any improvement to that bin can buy. The rows requiring per-cluster quantities are computed on the events whose energy key joins the geometry cache, 46% of the split; that matched subset is not representative — it scores 0.0427 against the full split's 0.0376 — so the shares and headrooms below describe the harder half and should be read as indicative rather than exact. "Sampling part" is $a/\sqrt{E}$ from a per-region fit of $\sigma_{\text{eff}}(E) = \sqrt{a^2/E + b^2/E^2 + c^2}$, the parameterisation of Section 5.1 written out, evaluated at the bin's median energy — the part no estimator can remove.

bin	share	bin σ_{eff}	aggregate if perfect	sampling part	headroom
60 mm low-E	11.3%	0.0513	0.0319	0.0225	large
40 mm low-E	9.6%	0.0583	0.0321	0.0469	small
30 mm low-E	6.7%	0.0699	0.0335	0.0513	small
15 mm low-E	3.8%	0.0675	0.0354	0.0417	moderate
120 mm low-E	2.0%	0.0540	0.0366	0.0291	moderate

Two of the three big bins are near their physics floor. At 40 mm the stochastic term alone is 0.0469 of the bin's 0.0583, and at 30 mm 0.0513 of 0.0699: 80 and 73 per cent of the width is sampling fluctuation of the shower itself, which is a property of the detector, not of the estimator. Only 60 mm low-E has real room — its sampling term is 0.0225 against a measured 0.0513 — and there the dominant term is $b/E = 0.0387$, an absolute error of $b \approx 0.29$ GeV at that bin's 7.5 GeV, which is the scale of the pileup fluctuation the window carries ($E_{\text{win}}/E_\gamma = 1.25$ at 60 mm). We tested the obvious handle: the residual there correlates +0.17 with the observable window energy, but removing a linear dependence on it, fitted on half the events and applied to all, makes every bin 7–10% worse. The window energy is already a global input; the network has consumed it.

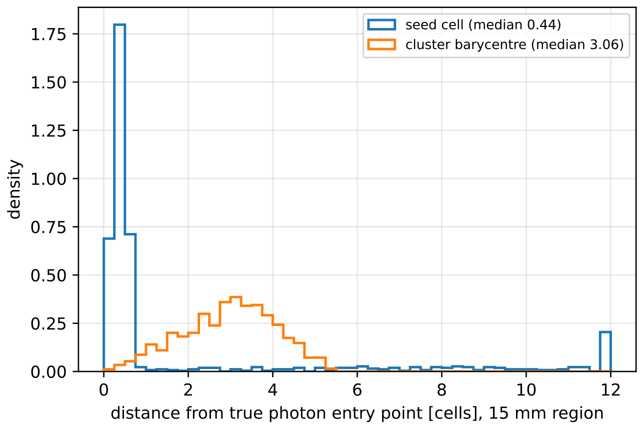

Figure 14: Distance of the two window-centre candidates from the true photon entry point, 15 mm region. The seed cell is the better photon locator by 7× in the median, yet centring on the barycentre gives the better resolution. The seed points at the photon but is wrong catastrophically in the tail (p90 = 8.3 cells), while the barycentre is wrong by a few cells almost always; with a window wide enough to absorb a few cells of error, the estimator that never fails badly is the better centre.

The variance component is exhausted. Across 28 single members the mean is 0.0394 and the best 0.0386; pooling all 28 gives 0.0376, only 4.7% below the member mean. The median member-to-member spread on a single event is 0.87% of the prediction against a total error of 3.76%, so the variance that ensembling removes is of order $(0.87/3.76)^2 \approx 5\%$ of the budget and most of it is already gone. Members disagree with each other far less than they disagree with the truth: the remaining error is common-mode.

Selection cannot help, by construction. 8.2% of events sit beyond $3\sigma_{\text{eff}}$ and carry 21% of the aggregate — perfect treatment of them alone would give 0.0296. The model flags them (AUC 0.93 from its own predicted width), but *removing* them does not narrow σ_{eff}: cutting to 80% efficiency on a width threshold fitted per region and validation-energy decile takes the aggregate from 0.0383 to 0.0412. σ_{eff} is the minimal 68.3% interval, and the flagged tails already lie outside it. Width-based flags are useful for analysis-level purity; they cannot improve a quoted resolution.

Two upper bounds are closed. A window centred on the true photon entry — unavailable at inference, measured as a diagnostic — is worth 4% on the aggregate, and the learned pointer takes most of it on the development split. A per-cell gate given truth is worth 10–12% more than one fitted from observables, and four supervision protocols failed to move the network toward it because the information is already reaching the answer by another route. Both of the mechanisms that a reader would propose first are therefore bounded, and the bounds are small.

What is left. Adding the pieces: the sampling floors of the two most populous weak bins are fixed by the calorimeter; the variance component is 5% of the budget and largely removed; selection is a null by construction; and the two structural bounds are worth a few per cent between them, most of it taken. Within this sample we estimate the reachable floor at $\sigma_{\text{eff}} \approx 0.038$, against the 0.0388 reported. The one lever measured to be worth more than a few per cent is the training-set size (Section 5.11), which is not a modelling choice at all.

5.6 Comparison of encoder architectures

The architecture campaign consumed the most compute and produced the cleanest null. Ten alternatives to the plain transformer were measured behind the same readout, loss and splits; several are re-implementations of published architectures and several were our own designs, built on physics arguments we found convincing at the time. None won.

The sharpest comparison is the paired one (Table 8): ParticleNet (EdgeConv with dynamic kNN) and GravNet re-implemented behind the identical readout, loss, splits and window, so the encoder is the only difference, with two seeds per baseline and five for the transformer. SpaceTformer performs best in every cell at every seed. Two caveats are required for fairness. The first is capacity: behind the identical readout, the transformer encoder holds 625k parameters against ParticleNet's 180k and GravNet's 279k, so it is the larger model by a factor 2.2–3.5 and the comparison is at equal *recipe*, not equal capacity. We did not grow the baselines to match, and the honest reading is that part of the margin may be capacity rather than architecture. What speaks against that reading is our own capacity sweep on the transformer itself (Table 10): d=256 with six layers scores 0.1179 and d=64 with two layers 0.0463, both far worse than the d=128 baseline, so on this sample parameter count is not a lever that buys resolution — which is also why the baselines were not given more. The second caveat: the baselines inherit our frozen training recipe, which was validated on the transformer; they received their published architectural defaults but no dedicated hyperparameter sweep, and the comparison should be read with that asymmetry in mind. A dedicated six-arm sweep (learning rate halved and doubled, and neighbourhood size k raised to 24, per baseline) closes this caveat, and we report it in full: every tuned arm still trails. ParticleNet reads 0.0456/0.0408/0.0422 (lr 10^{-4} / lr 6×10^{-4} / k=24) and GravNet 0.0443/0.0455/0.0427. The best of the six, ParticleNet at lr 6×10^{-4}, narrows the aggregate gap to +0.0009 (about twice the transformer's seed spread) and reaches 0.0785 at 15 mm low-E (within the transformer's spread there) — but remains +0.0045 behind at 30 mm low-E, five times the spread, at 3.6× the training cost. The robust statement is therefore: no tuned baseline matches the transformer at equal cost, and none matches it in the 30 mm weak bin at any cost we measured. The aggregate gaps of the paired default comparison (+0.0016 ParticleNet, +0.0038 GravNet against the transformer's five-seed mean 0.0399) exceed the transformer's own seed spread of 0.0005; the weak-bin gaps are larger still and consistent across bins, baselines and seeds.

Because only the encoder differs, the comparison can be paired event by event rather than read off seed spreads. Scoring every baseline seed against every transformer seed on the same events, and resampling events once per draw so both are scored on the same resample, the default ParticleNet is behind in at least 393 of 400 resamples in its worst seed pairing (ten pairings) and GravNet in 400 of 400 (ten pairings). Of the six tuned arms, five are behind in 400 of 400. The exception is the best of them, ParticleNet at lr 6×10^{-4}: its worst seed pairing beats the transformer in 46 of 400 resamples on the aggregate, so on the aggregate alone that single arm is not separated. It remains behind in the two weak bins (0.0809 against 0.0775 pooled over 15 and 30 mm low-E) at 3.6× the training cost. This is the sharpest form of the claim the evidence supports, and it is weaker than "wins everywhere" on the aggregate for that one arm.

Table 8: Head-to-head σ_{eff} under an identical fixed-split protocol. Single models, no ensembling, which is why the transformer row differs from Table 3's ensemble numbers; mean ± spread across seeds (five for SpaceTformer, two for each baseline). Training time is per model on one A100.

encoder	aggregate	15 mm low-E	30 mm low-E	time
SpaceTformer (ours)	$\mathbf{0.0399 \pm 0.0005}$	$\mathbf{0.0759 \pm 0.0058}$	$\mathbf{0.0768 \pm 0.0009}$	33 min
ParticleNet	0.0415 ± 0.0004	0.0815 ± 0.0003	0.0872 ± 0.0053	120 min
GravNet	0.0437 ± 0.0006	0.0861 ± 0.0045	0.0922 ± 0.0040	60 min

Table 9 is the full campaign. "Era" matters: early arms ran before the window and recentring fixes, against the matching reference of their time.

Table 9: All encoder families measured, fixed-split protocol, aggregate σ_{eff} with the era's matching transformer reference. Values are single-seed unless an arm was run with more, in which case the row is that arm's seed ensemble — the time-sliced row is a three-seed ensemble (0.0456 as a single-seed mean) while the Mixer row is one seed; the column is therefore not uniform in seed count and the rows should be compared with their own era reference rather than with each other. This is also why the ParticleNet/GravNet rows differ slightly from Table 8's two-seed means. The two time-factorised arms are one family measured in two eras. Screening = cut before a five-seed run.

encoder	lineage	aggregate	reference	outcome
pair transformer	pairwise tokens	0.0627	0.039–0.041	lost, early era
energy-flow network	Deep Sets / EFN [9]	0.0623	0.039–0.041	lost, early era
spacetime-factorised (v1)	TimeSformer [5]	0.0543	0.039–0.041	lost, early era
geometric pairwise-bias	ParT [4]	0.0420	0.0390	lost
grid CNN	dense image	0.0476	0.039–0.040	lost (2 seeds)
ConvNeXt token mixer	modern CNN	—	—	cut at screening
patch tokens (ViT-style)	spatial patching, cf. [6]	—	—	cut at screening
time-sliced attention (v2)	TimeSformer, on final config	0.0442	0.0402	lost
GravNet	[2]	0.0432	0.0402	lost, paired
ParticleNet	[1]	0.0412	0.0402	lost, paired
masked MLP-Mixer	[10]	0.0654	0.0402	lost
token transformer	[3]	**0.0390**		best encoder

Three failures are informative enough to deserve a sentence each. The *time-sliced attention* arm was our own flagship design — bucket cells by time pull, attend within buckets, exchange pooled summaries — built on the observation that pileup energy has a 13 ns time spread at 15 mm against 1.6 ns for the photon. It lost twice, on two eras of the pipeline, because slicing removes exactly the cross-time comparisons the full-attention model uses. The *grid CNN* represents the event as a dense image; it loses 20% to the token view of the same cells — under pileup, most of the grid is noise, and convolution must spend capacity learning to ignore what tokenisation simply omits. The *pairwise-bias* arm imports the Particle Transformer's physics-motivated attention bias, which helps on jets with hundreds of particles and well-defined pairwise invariants; at 81 noisy cells it only constrains the attention away from what the data wanted. The *masked MLP-Mixer* replaces attention with fixed learned mixing weights across token positions; it loses 63% (0.0654 vs 0.0402) because under pileup the informative cells move event by event — exactly the situation content-based attention handles and position-based mixing cannot. Capacity is not the explanation for any of this. Doubling the transformer to d=256/6 layers collapses under the frozen recipe (0.1179); we diagnosed the collapse directly, and it is an optimization artefact, not saturation: learning-rate warmup alone recovers to 0.0692, pre-layer normalisation [35] to 0.0474, and both together restore full trainability at 0.0412 — which still does not beat d=128's 0.0402. Halving to d=64/2 layers costs 15% (0.0463). The conclusion survives the diagnosis in a stronger form: the large model *trains* and still yields no gain. At 81–289 tokens per event, full attention is not the bottleneck, and no cheaper alternative matches it.

5.7 Interpretation of the learned gate

The readout's per-cell gate looks like it should estimate the photon fraction of each cell, yet on a paired overlay sample with per-cell truth labels its correlation with the true fraction is only 0.211. We measured four ways of "fixing" this: direct supervision of the gate on overlay truth, an auxiliary fraction head, per-cell prior features from a fitted estimator, and teacher distillation

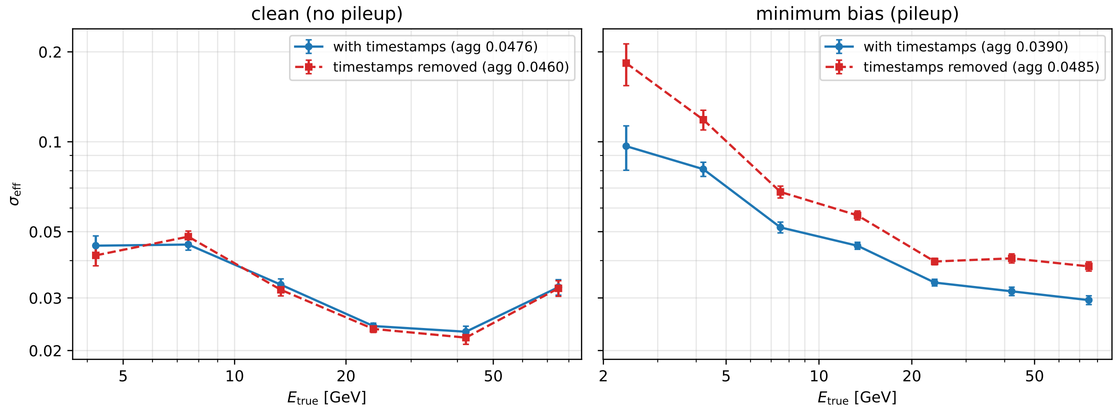

Figure 15: The timing ablation, side by side on the pileup-free clean sample (left; two-seed ensembles, earlier-era configuration) and the minimum-bias sample (right; the starting configuration). Removing the per-cell timestamps is neutral without pileup and costs 20% aggregate — concentrated at low energy — under pileup: timing is a pileup-separation tool, not an energy observable.

from that estimator. All four are neutral or worse. Meanwhile a gradient-boosted per-cell estimator given the identical per-cell observables reaches correlation 0.945 on the same sample — the information is present in the inputs, and the network routes it around the gate rather than through it. The gate is a learned attention weight inside an integral, not a physical quantity, and constraining it to be physical costs resolution. We flag this because the "interpretable gate" framing is publication-ready and, on this measurement, wrong.

The gate study sat inside a larger arc on *overlay supervision*: manufacturing per-cell truth by overlaying clean photon showers onto pileup and training against those labels. Its structural half worked — an explicit subtract-then-calibrate decomposition, where one branch estimates the pileup field and a second calibrates what remains, reached 0.0465 as an ensemble in its era — but the supervision half did not: training the subtraction branch on overlay labels was neutral against letting it learn freely. A domain measurement explains why: synthetically overlaid events are distinguishable from real minimum-bias events by a classifier at AUC 0.93 on cell-level observables, so labels manufactured by overlaying do not come from the distribution the model must serve. Structure helped; manufactured labels did not.

5.8 Timing

Per-cell timing is the physics handle expected to matter under pileup, and Figure 15 shows it is a *pileup* handle specifically. Retraining the starting configuration with both timestamps removed gives aggregate 0.0485 against 0.0390, 15 mm low-E 0.2033 against 0.1654, and 30 mm low-E 0.1556 against 0.0957 — timing carries 20% of the aggregate resolution and up to 38% in the weak bins — while the identical ablation on the pileup-free clean sample is neutral (0.0476 with vs 0.0460 without). The timestamps carry no energy information; their entire value is separating overlapping sources in time

We could not obtain the simulated timestamp resolution from the production configuration, so we measured it on the samples themselves, taking the energy-weighted time of a cluster's top-decile cells as the reference. It is strongly energy-dependent: on the clean sample the per-cell spread falls from 0.756 ns at 2–10 MeV to 0.435, 0.263, 0.156 and finally 0.038 ns above 3 GeV, a factor twenty across the range, and every bin degrades by 1.7–2× under pileup (1.275 → 0.132 ns over the same cells). A cell's timestamp is informative only when the cell is energetic — which is why probe (iii)'s "quarter of the per-event spread" is a tight constraint in the cells that carry the photon and a loose one in the low-energy cells where the pileup sits, and why an explicit time gate and a resolution-weighted pull both added nothing. Every *engineered* use of timing loses to handing the network the raw timestamps: three time-pull coordinate constructions (per-cell time scaled by its expected spread), hard timing cuts, time-sliced attention, an inverse-variance front/back combination, and a Fourier basis on the raw time channels (a null: 0.0390 against the era's 0.0391 at three seeds) were all measured and all reduce or at best match. The network already extracts what the timestamps carry; pre-digesting them removes information. The result independently supports the HL-LHC timing programme — per-cell timing in high-granularity calorimeters [15] and dedicated timing layers for four-dimensional vertexing [32]: per-cell timestamp resolution, and the fraction of cells that carry a timestamp at all, feed a model like this one directly.

Three interventional probes on the trained model say *how* the timestamps are read, which the retraining ablation cannot. These perturb the inputs of one trained member at evaluation and recalibrate per variant, so they are scored against that member's own unperturbed 0.0402 and not against the ensemble of Table 3; a model asked to read corrupted timestamps it was trained to trust fails far worse than one retrained without them, which is why the numbers below are an order of magnitude larger than the 0.0485 of the retraining ablation. (i) Permuting the timestamps among the cells of the same event — the identical multiset of values, reassigned — degrades the aggregate to 0.398, *worse* than zeroing them (0.310), so the model reads per-cell time structure rather than any pooled event statistic; (ii) a rigid shift of every timestamp in an event, which changes nothing pairwise, already costs 44% at half a standard deviation — the model anchors on the absolute median-subtracted scale where in-time means near zero, which together with the pairwise-bias encoder's null (Section 5.6) closes the case for an explicit relative-time attention mechanism; and (iii) coarsening the timestamps onto a grid costs 11% at a quarter of the per-event time spread and 19% at half — the network exploits timing down to roughly a quarter of the spread the sample provides, so the simulated timestamp resolution matters at exactly that grain.

5.9 Negative results

Table 10 is the record of measured null results beyond the encoder campaign. Each row is a mechanism a reviewer — or we ourselves — would otherwise propose. For scale: the single-member spread on the aggregate is 0.0010 and the bootstrap error on the ensemble aggregate is 0.0002 (Section 3); rows marked "worse" exceed it, rows marked "neutral" sit within it.

Two of the new rows carry lessons beyond their null. The quality-selection row fails *by construction*, not by accident: the predicted interquartile width does flag the catastrophic events, but σ_{eff} is the minimal 68.3% interval — a robust statistic — and the flagged tails already lie outside it, so removing them cannot narrow it (cutting to 80% efficiency *worsens* the aggregate from 0.0383 to 0.0412; the thresholds were fitted per region and validation-energy decile to exclude the trivial version of the cut that simply empties the hard bins). Width-based flags remain useful for analysis-level purity; they cannot improve the quoted resolution. And the train-time augmentation null is a control for the scaling claim: an eight-fold increase in geometric variety yields nothing, while the measured curve implies that tripling the *real* sample yields $\sim$18% — what $N^{-0.18}$ purchases is diversity of pileup configurations, which symmetry transforms cannot manufacture. The data request has no augmentation shortcut.

Beyond these, fifteen per-cell and global feature mechanisms were measured on the same protocol. The ones that stayed: extra reconstruction globals, local density, and the clean-sample auxiliary. The ones measured neutral-or-worse, each with its row in the repository ledger: time-pull coordinates (+0.017 at 15 mm low-E), orthogonal-residual coordinates, absolute time, local energy fraction, depth ratio, gate-side global injection, physics/occupancy summaries, FiLM conditioning on pileup context, signed gates (+0.009 at 15 mm low-E on the recentred configuration), trainable gate temperature, low-energy cell reweighting at four strengths, and millimetre-unit geometry (neutral on the final configuration; its one gain before recentring was

Table 10: Negative results, all measured under the frozen protocol. Aggregate σ_{eff} comparisons are against the matching-protocol value 0.0379 (the half-sample CV ensemble current when each was measured) or 0.0402 (fixed-split single seed) as labelled.

family	protocols tried	outcome
gate supervision	4 (direct, aux head, prior feature, distill)	neutral or worse
engineered timing	5 (pulls $\times$3, cuts, slices, front/back comb.)	all worse
training recipe	lr $\times$2, batch $\times$2, jitter $\times$2	0.0422–0.0462 vs 0.0402
ensembling	learned stacking on held-out data	0.0414 vs 0.0379, worse
recalibration	post-hoc per (region $\times$ E-decile)	0.0386 vs 0.0379, worse
test-time augmentation	D4 averaging	neutral, 8$\times$ cost
capacity	d=256/6L; d=64/2L	0.1179 / 0.0463, both worse
capacity, diagnosed	+ warmup; + pre-LN [35]; + both	0.0692; 0.0474; 0.0412 — trains, no gain
time representation	Fourier basis on raw time channels	0.0390 vs 0.0391 (3 seeds), neutral
slot decomposition	K=4 competing readout queries	0.0406 vs 0.0402, neutral
time-gated readout	multiplicative time gate on the final window	0.0399 vs 0.0399 mean, neutral
train-time augmentation	D4 transforms of the window	0.0401 vs 0.0402, neutral
quality selection	cut on predicted quantile width	worse at every efficiency (see text)
core-restricted sum	gated sum over w=4; w=2 core	0.0404; 0.0436 vs 0.0399
core sum on a pointed window	w=2 core, learned centre	0.0389, equal to full sum

superseded by recentring itself). The consistent reading across all fourteen: the token set already carries the information, and the model finds it without hand-built coordinates.

5.10 Computational cost

A reconstruction algorithm is only deployable at a cost, so we benchmark inference throughput on commodity hardware (laptop CPU, i9-13900HX, batch 64, median of 5 timed passes on an idle machine). The transformer carries 623,238 parameters against the GBDT's 18,300 tree nodes. Figure 16 places each model family on the accuracy–throughput plane. At the 9×9 window a single transformer runs at ∼5,000 clusters/s (0.20 ms per cluster) and the five-seed ensemble at 992 clusters/s; the GBDT reference is 3.3× faster per cluster than a single transformer pass and 3.15× worse in resolution; the analytic calibrated sum is effectively free (5.6×10^6 clusters/s) and 4.6× worse. The final 17×17 recentred configuration pays the quadratic attention cost of 289 vs 81 tokens: 555 clusters/s per model, 111 clusters/s for the five-member ensemble, measured on the same harness. On one H100 GPU the same configuration reaches 1,017 clusters/s at batch 1 (latency-bound) and 41,345 clusters/s at batch 1024 — 24 μs per cluster for a single model, 121 μs for the five-member ensemble — which brings batched inference into the regime where a conversation about high-level-trigger deployment becomes quantitative (`reports/benchmark_gpu.csv`). Accuracy per compute is therefore a real trade-off the experiment can choose along: from the free analytic sum to the full ensemble spans four orders of magnitude in throughput for a 4.7× span in resolution.

5.11 Dependence on training-sample size

With representation, architecture, supervision, recipe, ensembling and calibration each closed by measurement, the remaining open lever is training-set size, and we measured its slope directly: the final configuration retrained on deterministic 25%, 50% and 75% subsamples of the training set (validation and test untouched, same recipe, three seeds per point) plus the full-data point (Figure 17). The four points are strikingly linear in log–log and give

$$\sigma_{\text{eff}} \propto N^{-0.18},$$

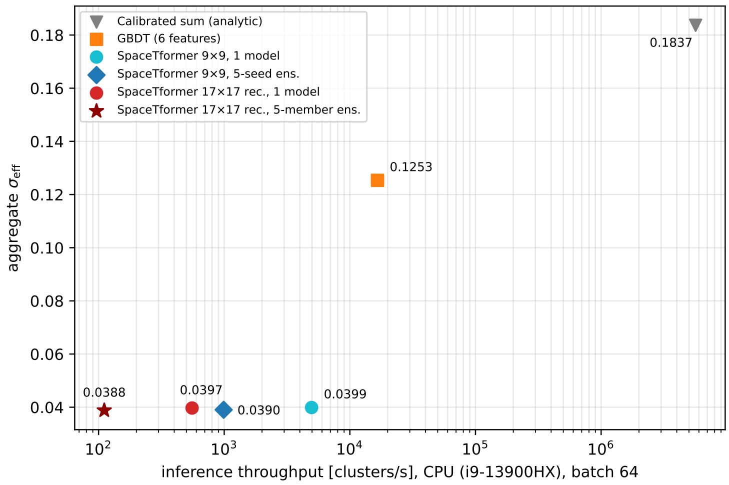

Figure 16: Aggregate σ_{eff} against inference throughput (CPU, batch 64; log scale). The four left points use the fixed-split protocol; the two 17×17 points use the single-member mean (0.0397) and the cross-validated ensemble (0.0388), with the forward-pass rate measured at 289 tokens on the same harness. The transformer obtains its resolution by reading cells individually, and the window that fixed the weak bins costs a further 9× in attention compute.

identically for single models (0.0524 → 0.0466 → 0.0439 → 0.0402) and for three-seed ensembles (exponent −0.179). The worst bin does not merely follow: 15 mm low-E runs 0.1254 → 0.1017 → 0.0956 → 0.0729 across the same points, an exponent of −0.35, about twice as steep as the aggregate. More data helps the bin that needs it most, faster than it helps the average. This within-protocol curve supersedes two earlier confounded two-point estimates (−0.13 and −0.28) and lands between them; power-law scaling of error with dataset size is the generic empirical behaviour of deep networks [34, 33], so the linearity is expected even if the exponent is task-specific. Extrapolated threefold — an extrapolation, and labelled as such — it predicts an aggregate of ≈ 0.033. The qualitative case is independent of the fit: every intervention that did not add training data saturated at aggregate ≈ 0.039, and the residual error is dominated by per-event pileup magnitude (Section 5.4), a statistical quantity that more data smooths.

6 Limitations and future work

Five limitations bound the claims. (0) The clearest one is recent: the two-stage window's development-split gain does not reproduce under cross-validation (Section 5.3), which is a warning about every development-split number in this paper. They are labelled wherever they appear, but a margin of a few per cent measured on the fixed split should be treated as a hypothesis until the ten-fold protocol confirms it. (1) Everything is measured on one simulated sample of one detector concept; the diagnosis method transfers, the numbers may not. Two internal transfers partially mitigate this: the pileup-free clean sample is a second distribution on which the timing conclusion reproduces with the opposite sign (Figure 15), and the window diagnosis holds across five detector regions whose granularity spans a factor of eight — effectively five related sub-detectors. (2) The architecture comparison is paired and now includes a per-baseline hyperparameter sweep (learning rate, neighbourhood size), but the sweep is small; its conclusion is about this task's regime, not about the architectures in general. (3) The scaling prediction at 3× data is an extrapolation beyond the measured range of the (otherwise clean) four-point curve. (4) The half-sample preview of the primary table read 0.0379 aggregate; the complete ten-fold table reads 0.0388 — a real lesson in why half-done cross-validation must not be quoted as final, which we record here deliberately.

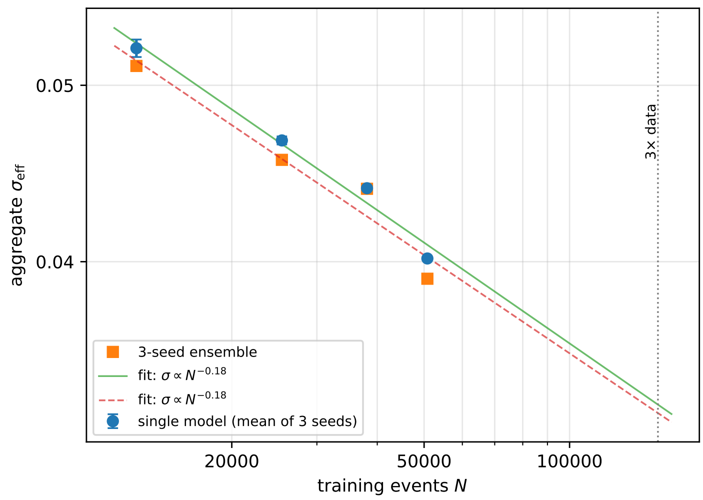

Figure 17: The measured data-scaling curve: aggregate σ_{eff} of the final configuration retrained on 25/50/75/100% of the training set (fixed-split protocol, three seeds per point; the 75% point completed one seed). Both the single-model mean and the three-seed ensemble fit $\sigma_{\mathrm{eff}} \propto N^{-0.18}$; the dotted line marks a tripled sample, at a predicted aggregate of ≈ 0.033.

On venue fit, we state the contribution class plainly: this paper offers a measurement-driven diagnosis, a controlled architecture comparison, and a ledger of verified negative results — not a new architecture; it should be evaluated as such.

Three directions follow directly from the measurements. The catastrophic tails (Section 5.4) are a reconstruction problem — wrong-photon matches and overlapping clusters — addressable upstream of any regressor. The timing-coverage correlation suggests that denser timestamp coverage is a detector-design lever this model would convert directly into resolution. And the data request is now quantitative: tripling the minimum-bias simulation is predicted, with stated caveats, to bring the aggregate below 0.035.

7 Conclusions

On a simulated minimum-bias PicoCal sample, SpaceTformer reaches $\sigma_{\mathrm{eff}} = 0.0388 \pm 0.0002$ aggregate, fully out-of-sample over the complete ten-fold protocol — a factor 3.15 ± 0.05 over a six-feature GBDT reference on the events the two share — and beats paired re-implementations of ParticleNet and GravNet in every measured cell at every seed, at a fraction of their cost. The improvement over our own starting point — 53% in the worst region–energy bin — came entirely from measuring and fixing the input representation: window containment and window centring. The same diagnosis, pushed one step further, produced the learned two-stage window of Section 5.3, which points before it measures and takes the development-protocol ensemble to 0.0378; a cross-validated re-run of the same construction does not reproduce that margin, and we report both. What does hold under either protocol is the auxiliary position head, at about 1%. Everything else we tried is reported with its numbers in Sections 5.6–5.9 and the scored ledger versioned with the code, so it does not have to be re-tried silently. Pricing the residual bin by bin (Section 5.5) puts the reachable floor on this sample at $\sigma_{\mathrm{eff}} \approx 0.038$: the two most populous weak bins are 73–80% sampling fluctuation, the variance ensembling removes is 5% of the budget and largely gone, selection cannot narrow a robust interval, and the two structural bounds a reader would propose first are worth a few per cent between them. The measured scaling curve, $\sigma_{\mathrm{eff}} \propto N^{-0.18}$, therefore makes the path below that a data request rather than an architecture search.

Acknowledgements

This work was carried out within Google Summer of Code 2026 for the LHCb PicoCal project. I thank my mentors Felipe Luan Souza de Almeida and Carla Malin Benito for the simulated samples, weekly guidance, and the error-analysis discipline this paper is built on. [LHCb publication-approval status to be settled with the mentors before any submission; single-author use of collaboration simulation may require Editorial Board review or release as a collaboration note.]

Code and data availability

All experiments run from a single training script with a flag for every mechanism in this paper; the prediction CSVs behind every reported number and the scripts that generate every figure are versioned alongside the code at https://github.com/Lworakan/GSoC2026-picocal-spacetime-transformer, together with a locked environment and a test suite whose regression canary is σ_{eff} itself.

The script's defaults are the starting configuration, not the primary one, so the three member families of Table 3 are named explicitly. For fold $k \in \{0, \ldots, 9\}$:

```
train_picocal.py --window 8 --extra --dens --recenter --cleanaux \
                 --aux --fold k --nfold 10 --seeds 0 1 2
train_picocal.py --window 8 --extra --dens --recenter --cleanaux \
                 --rc-regions 0 1 --fold k --nfold 10 --seeds 0
train_picocal.py --window 4 --extra --dens --recenter --cleanaux \
                 --allcells --fold k --nfold 10 --seeds 0
```

The five members of a fold are combined by per-event median; the ten folds are concatenated. One gap should be stated plainly: the trained weights behind the primary result were produced on rented cloud hardware and are not in the repository, so a reader can re-score our predictions exactly but must retrain to reproduce them from scratch.

The simulated samples belong to the LHCb collaboration and are not redistributed.

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